

Network Design Report

Prepared for
TerraGaff Ltd

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Abstract

This is a report proposing professional network design solutions of TerraGaff Ltd., a mid-sized company that is moving to a newly edifice for offices consisting of three stories, located in Ulverston, Cumbria. Each floor has different connectivity and service needs, such as secure internet access, internal services, and wireless connectivity among staff as well as clients. Among these three options of design, a Layer 3 VLAN-based network design has been suggested. It has all benefits of proper segmentation, scalability, and better security of employee, server, and guest networks. The solution articulated above also takes care of clear IP addressing mechanism, dynamic addressing through DHCP, hosting of web and email services from the internal network, and modern wireless setup. The part will balance performance, security, and costs in serving today's operations and growth in future. Roughly £18,450 is estimated for implementation costs to cover hardware, cabling, and basic installation. Cyber security measures of the best practices in the industry (network segmentation, WPA3 wireless encryption, and firewall integration) are included in the plans to protect the assets and data of the company. This design allows TerraGaff Ltd to shift into a new location and ensures the uninterrupted delivery of similar services at the new location.

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1 Company and Network Requirements

1.1 Company Overview

TerraGaff Ltd is a mid-sized enterprise in the process of relocating to a new three-floor office building in Ulverston, Cumbria. The company's operations are divided by floor, with administrative activities located on the first floor, and client-facing or service functions located on the second and third floors. The new office will be connected to the internet via a local ISP, Connexions.

1.2 General Network Requirements

Based on the case brief, the following general requirements have been identified:

- Each of the three floors must operate on a separate subnet to enable logical segmentation.
- Wired connections will be used to interconnect the three subnets.
- One wireless access point (WAP) is required per floor to provide Wi-Fi coverage.
- Devices (wired and wireless) must receive IP addresses dynamically via DHCP.
- The network must support the hosting of an internal company website and an email service under the company's own domain, using in-house servers.
- The router connecting to the ISP and the core switch will be located on Floor 1, along with the company's servers.
- Printers must be accessible via the network: two on Floor 1, one on Floor 2, and two on Floor 3.
- Guest wireless access must be available on each floor, with Floor 1 allowing for 10 guests and Floors 2 and 3 each supporting several guest devices equal to the number of employees on that floor.

1.3 Floor-Specific Requirements

Floor	Purpose	Employees	Wired Devices	Wireless Devices	Guest Wi-Fi	Printers
1	Administration	20	20	40	10	2
2	Client Services	30	30	60	30	1
3	Client Services	50	50	100	50	2

2 Additional Considerations and Cybersecurity Requirements

In addition to the technical and structural network requirements within the scope of this brief, critical cybersecurity- and operation-related considerations must be accounted for the network to be secure, scalable, and efficient for TerraGaff Ltd. These provisions have been put in place following best practices in network design and current cybersecurity standards. Oftentimes, a human is the weakest link in cybersecurity (ICO, 2023).

2.1 Security Measures

a) Network Segmentation and VLANs

VLANs are additionally proposed to logically segment different forms of traffic: staff, guest, printers, and servers. By minimizing the attack surface through limited broadcast domains and access enforcement at the switch level, the security shall be enforced. Zero Trust Architecture (ZTA) model should also be considered by TerraGaff Ltd, requiring continuous verification of user and device identities irrespective of their position within the network, so that strict access controls can be established in case of lateral interactions (Cisco, n.d.).

b) Firewalls and Intrusion Detection/Prevention

Next-Generation Firewall (NGFW) must be implemented between ISP router and internal network for the deep inspection of both inbound and outbound traffic. Integrated IDS/IPS would permit TerraGaff to detect anomalies or malicious activities at such an early stage that remediation processes would still prove highly effective.

c) Encrypted Guest Wi-Fi

Guest access should use VLAN isolation and captive portals to ensure no access to internal systems.

d) Wireless Network Security

Every WAP should be outfitted with WPA3 at a minimum for all possible wireless traffic. Then an Enterprise-grade access control should be employed using 802.1x with RADIUS for internal user access while a simpler approach with either a Captive Portal or Pre-shared Key can be used for guest access.

e) Endpoint Protection

All user endpoints and installations should be configured for a centralized antivirus/EDR solution and patched regularly to ensure that they remain relevant against emerging vulnerabilities.

f) Data Backup and Disaster Recovery

In-house servers (mostly email servers and web servers) must have daily database backups to both local storage and encrypted cloud space to maintain data integrity during user failure and ransomware assault.

g) Physical Security

Therefore, it is best for servers and networking equipment to be placed in a server room, which should be locked with limited access, and monitored by CCTV, with uninterruptible power supply (UPS) units in place to keep the up time during power outages.

2.2 Scalability and Maintainability

This means that the future growth of the network should not be hindered. More users, devices, or services will be accommodated without the necessity for complete redesign. Modularity provided by switches, structured cabling, centralized-management platforms, etc., will help in bringing down long-term operation costs.

2.3 Legal and Compliance Factors

TerraGaff must comply with UK-GDPR, securing personal data in transit and at rest, with regular audits and policy reviews.

3 Network Design Options

We examined into three potential designs, each of which provided a different answer to the technical and business needs of TerraGaff Ltd.; each design describes its way of addressing connectivity, scalability, and security.

3.1 Option A: Basic Layer 2 Switched Network

This type of flat network would use unmanaged switches connected to one subnet per floor, with a basic stand-alone router doing DHCP and NAT. Wireless access points would be installed with just a basic level of WPA2/WPA3 encryption. The servers reside on the same LAN segment.

Advantages:

- Inexpensive equipment
- Simple installation
- Configuration needed is minimal

Disadvantages:

- There is no logical segmentation, except for the physical subnets.
- Bad scalability and security.
- Single point of failure (i.e. router)
- Difficult to segregate traffic (guest vs internal)

3.2 Option B: Layer 2 + Layer 3 Segmented Network with VLANs and Managed Switches.

Every floor will have its subnet, with managed switches talking and allowing the different VLANs to segment users, such as guests, servers, and printers, maintaining a Layer 3 switch or router-on-a-stick to perform inter-VLAN routing. Setting up hosting services for DHCP and DNS are in full swing. Wireless networks are VLAN-aware (SSIDs mapped to VLANs).

Advantages:

- Good logical separation of network roles (users, servers, guests, etc.)
- Improved traffic control, security, and scalability
- Central management
- Firewall rule easier to apply per VLAN

Disadvantages:

- It increases equipment and configuration costs.
- It needs well-trained IT personnel for setup and maintenance.

3.3 Fully Converged Network with Cloud-Based Management

Description:

Cloud-based management of switches, access points, and firewalls (e.g., Cisco Meraki, Ubiquiti UniFi). A centralized dashboard can manage VLANs, wireless SSIDs, firewall

policies, and user analytics. Keep servers on-premises, but cloud control allows remote management.

Advantages:

- Simple centralized management via the cloud portal
- Automatic updating of firmware/software/security
- Analytics and traffic shaping
- Scalability and ease of use

Disadvantages:

- Significant initial and recurring subscription costs
- Vendor lock-in
- Management requires reliable internet connectivity

3.4 Recommended Design

The recommended plan for TerraGaff Ltd. is Option B. A Layer 2 + Layer 3 Segmented Network with managed VLANs and switches, as it offers great advantages in security, scalability, and cost. This option provides total isolation of the guest and internal networks, keeps secure internal services such as e-mail and web hosting, and is a structure that can easily be upgraded as the organization grows. Although Option C would allow a more manageable connectivity system, the associated costs and probable cloud connectivity requirements may override their benefits in the present situation of a rather matured existing infrastructure for TerraGaffs. On the contrary, Option A does not meet either of security or scalability concerns.

4 Detailed Design Proposal

As justified by Section 4, this proposal designs a Layer 2 + Layer 3 network with VLAN segmentation and managed switches supported by central DHCP and DNS services. This design meets current operational demands at TerraGaff and leaves room for growth possibilities.

4.1 IP Addressing and Subnets

For network segmentation, they will use the Class B private IP address range of 172.16.0.0/16 about flexibility and scalability.

Floor	Subnet (CIDR)	Range	Devices Required	Reserved Hosts
1	172.16.1.0/24	172.16.1.1–254	~70	254
2	172.16.2.0/24	172.16.2.1–254	~120	254
3	172.16.3.0/23	172.16.3.1–510	~200	510
Servers	172.16.10.0/28	172.16.10.1–14	~10	14
Guests	172.16.20.0/22	172.16.20.1–1022	up to 90+ users	1022

- Each subnet will have DHCP giving IPs while static IPs will be for infrastructure.
- VLANs will map to each subnet: e.g. VLAN 10 for Admin, VLAN 20 for Floor 2, VLAN 30 for Floor 3, VLAN 99 for Guests.

4.2 Equipment Specifications

Component	Specification / Model	Quantity
Core Router / Firewall	Ubiquiti EdgeRouter 4 or Cisco ISR 1100	1
Layer 3 Managed Switch	Cisco CBS350-24T (or equivalent)	1 (core)
Floor Switches	Cisco CBS250-24P or TP-Link JetStream	3 (1 per floor)
Wireless Access Points	Ubiquiti UniFi 6 Lite or Cisco WAP581	3 (1 per floor)
Servers (on-premises)	2U Rackmount (Xeon, 32GB RAM, RAID storage)	2
Printers	Network-ready LaserJet	5
Patch Panels & Cabling	Cat6, shielded	As needed

4.3 Servers and Internal Services

Server	Purpose	OS / Software
Web Server	Host company website	Ubuntu Server + Apache/Nginx
Mail Server	Handle internal/external mail	Postfix + Dovecot + TLS
File Server	Shared storage and backups	Samba / Nextcloud
DHCP/DNS	Central IP and name service	ISC DHCP + BIND
Monitoring	Network health checks	Zabbix / Nagios

Backups will be configured for daily sync to an encrypted external NAS and off-site cloud solution.

4.4 Network Diagram

First a draft diagram was sketched on paper because this way is better to put your thoughts in order and organize the overall network. Then the actual and final network diagrams were created to accurately represent each floor's infrastructure.

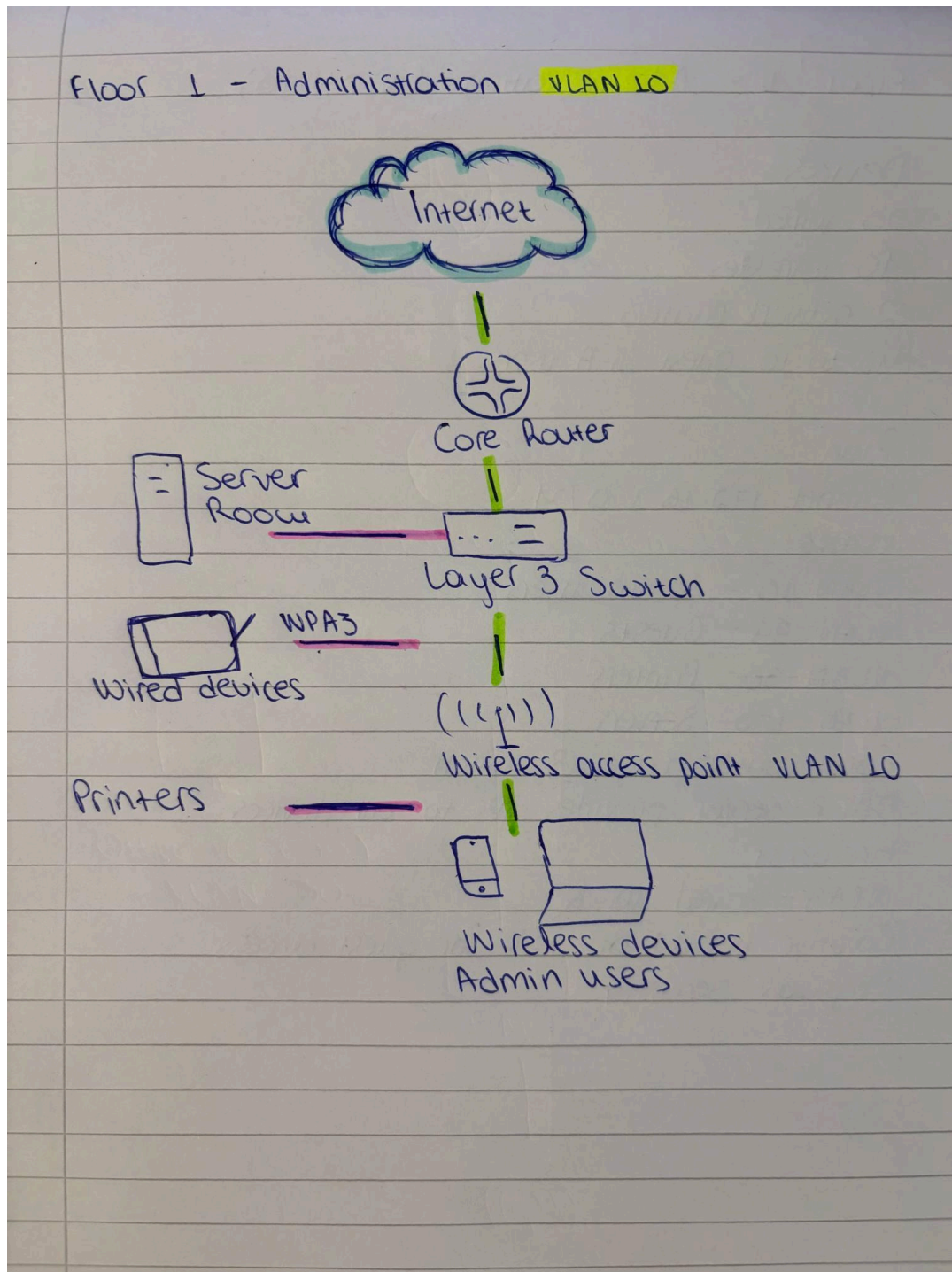


Figure 1 Floor 1 - Draft diagram

Floor 2 - VLAN 20

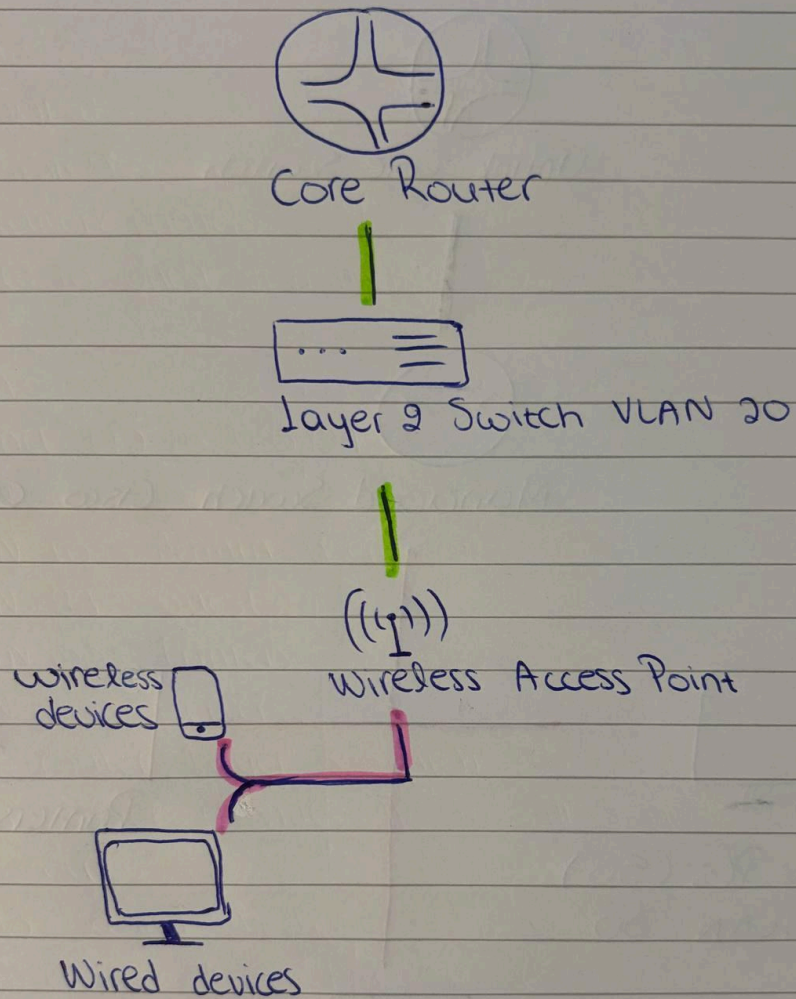


Figure 2 Floor 2 - Draft diagram

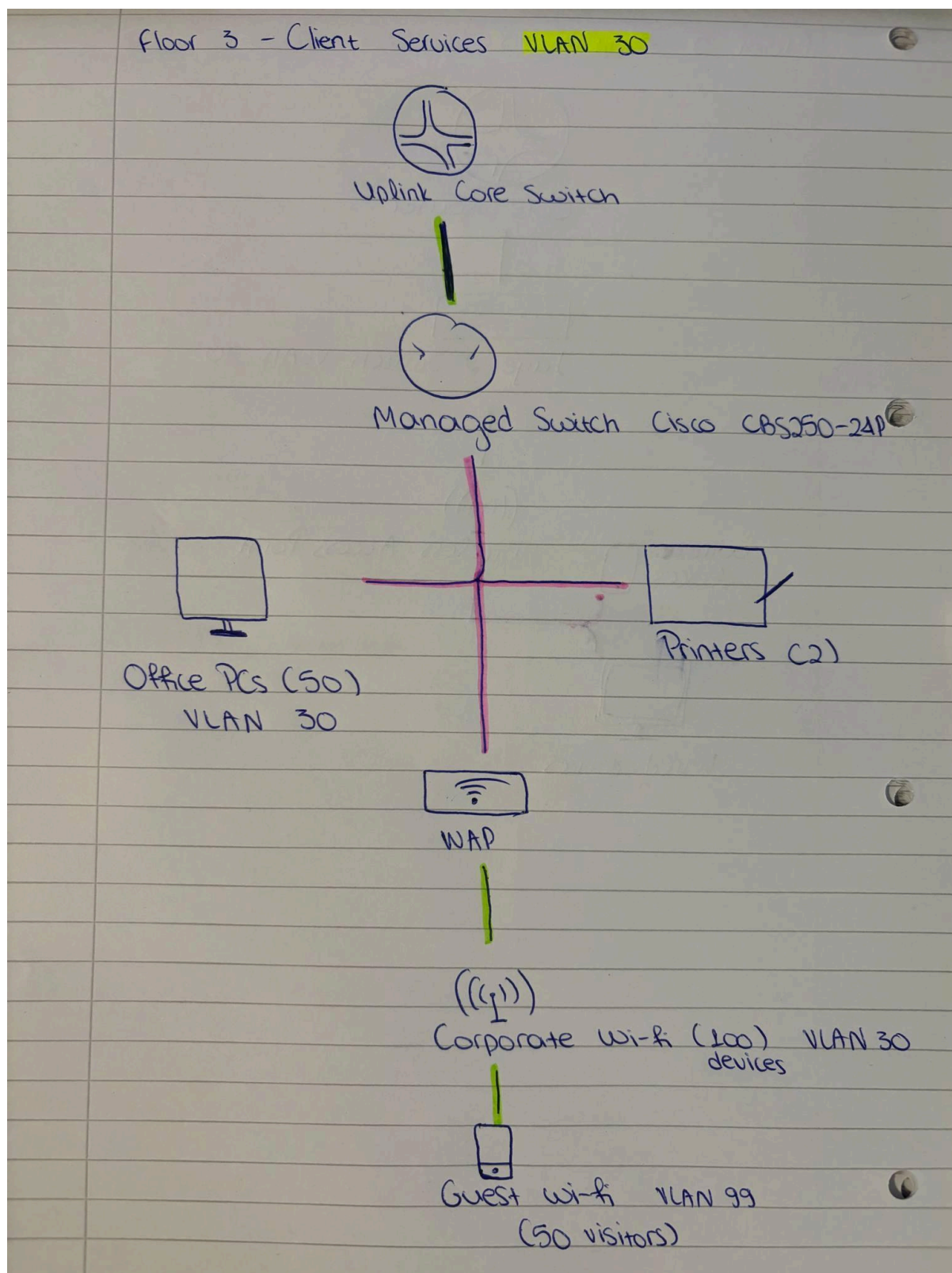


Figure 3 Floor 3 - Draft diagram

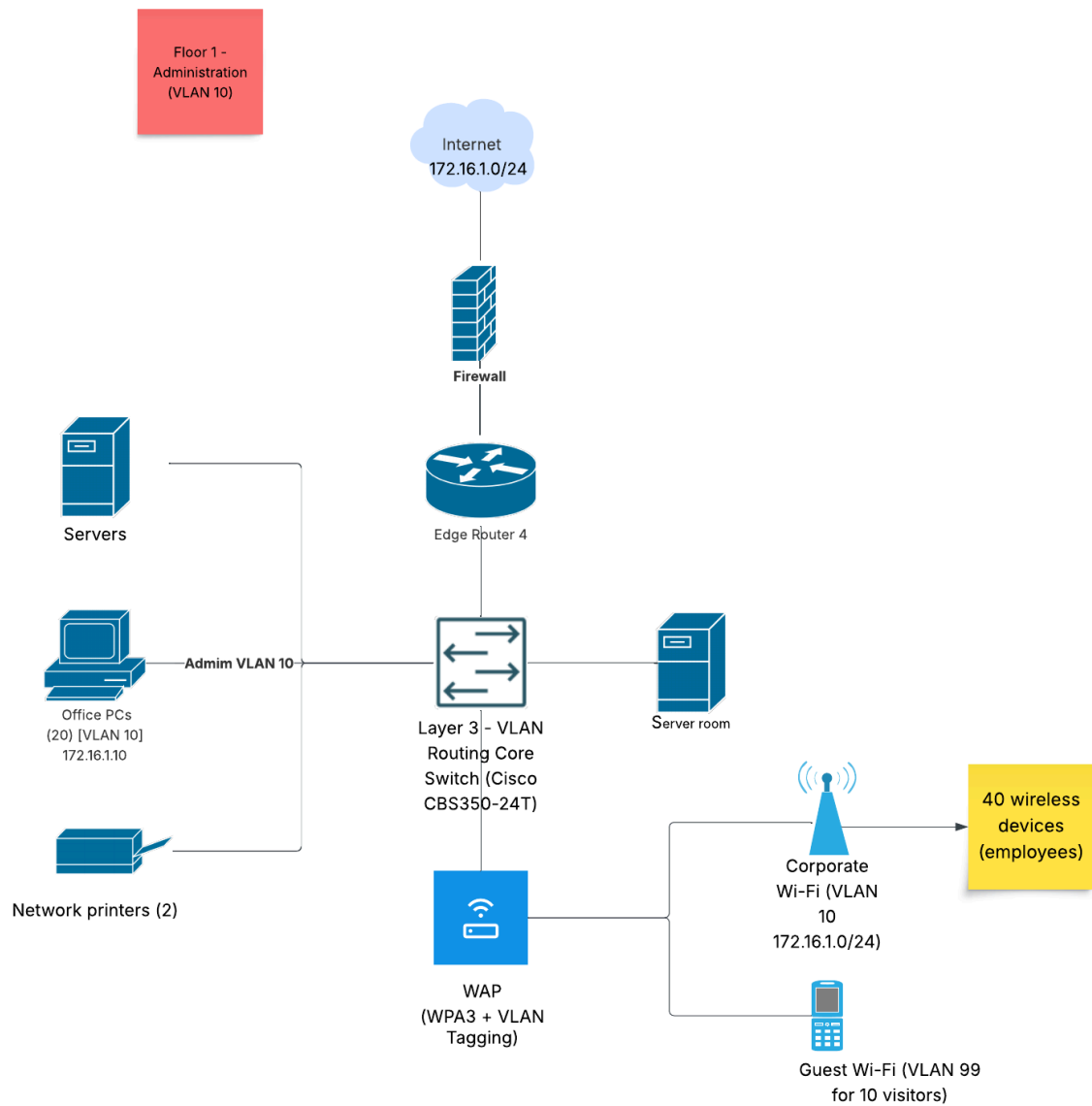


Figure 4 Floor 1 - Final diagram

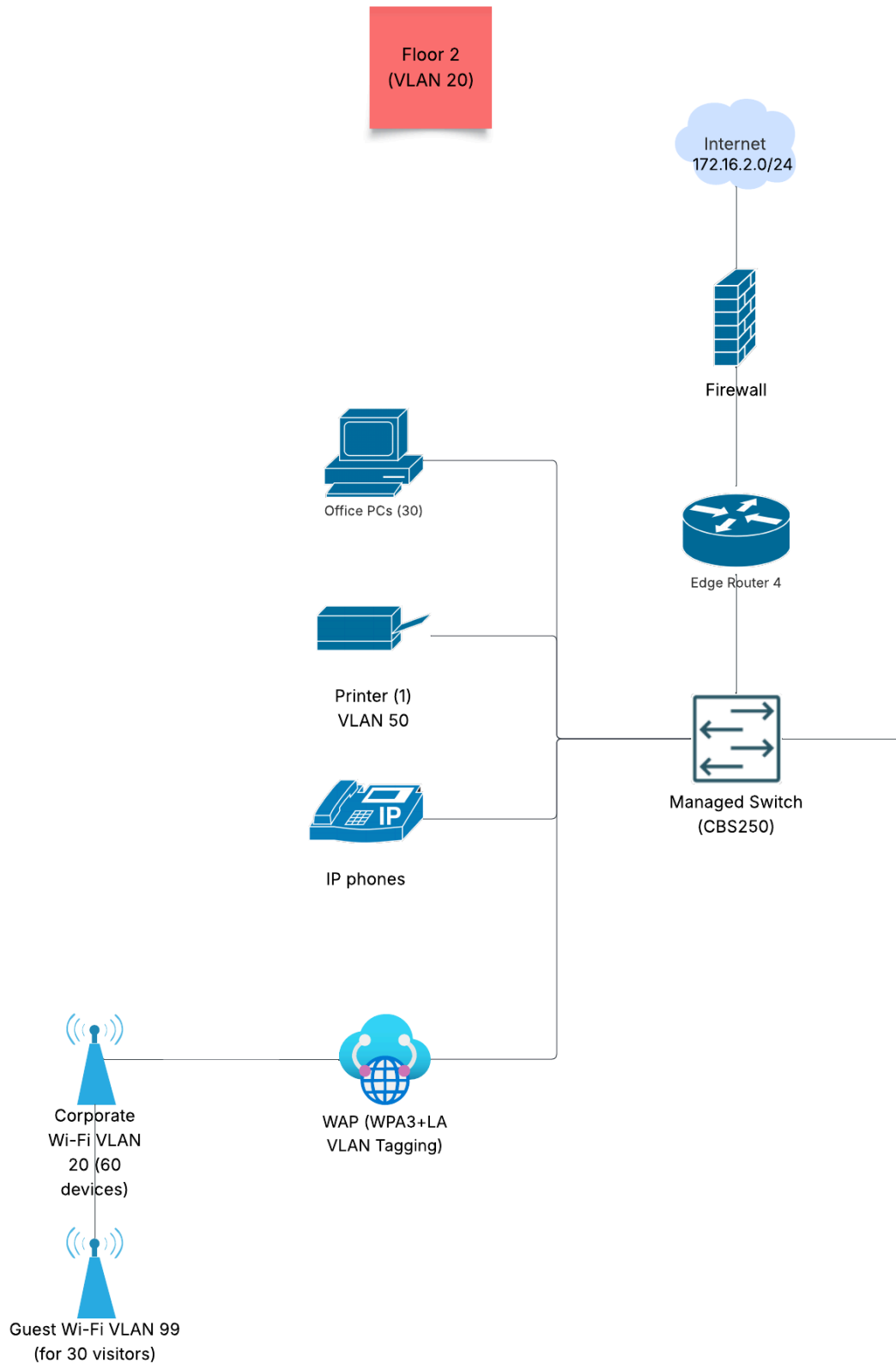


Figure 5 Floor 2 - Final diagram

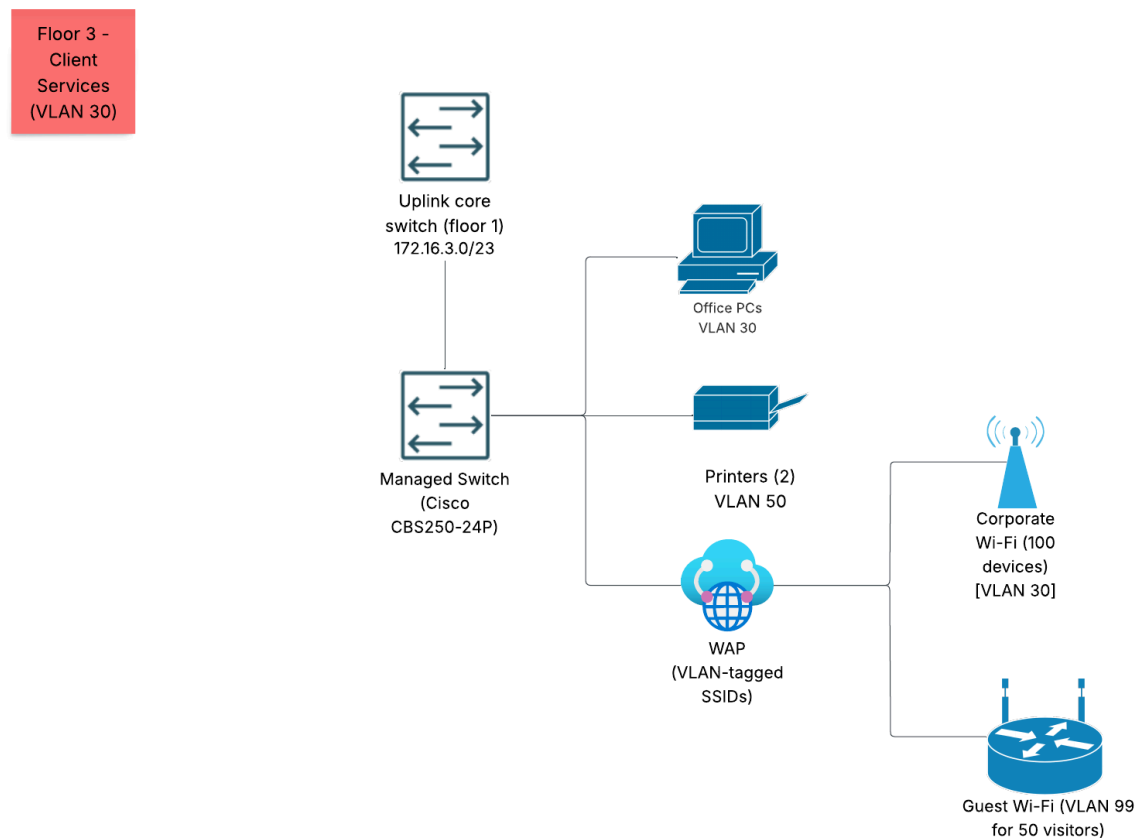


Figure 6 Floor 3 - Final diagram

4.5 Management and Monitoring

All switches and APs will support the SNMP and/or Syslog mechanisms used for monitoring. The administrator will be granted access to all devices through a dedicated management VLAN.

Security rules shall be enforced via:

- Inter-VLAN firewall rules (such as guest VLAN cannot access server VLAN)
- Port-based MAC filtering on switches
- Secure SSH/HTTPS access for management only from trusted subnets

5 Cost Estimate

This section outlines the estimated costs associated with the recommended network design for option B. While precise figures may vary based on vendor agreements and procurement options, the following estimates are based on current UK market pricing and installation assumptions.

5.1 Equipment Costs

Component	Unit Price (est.)	Quantity	Total (£)
Ubiquiti EdgeRouter 4	£180	1	£180
Cisco CBS350-24T Switch	£320	1	£320
Cisco CBS250-24P Switch	£250	3	£750
Ubiquiti UniFi 6 Lite AP	£115	3	£345
Rackmount Servers	£1,200	2	£2,400
Network Printers	£150	5	£750
Patch Panels & Racks	£200	1	£200
Firewall License/Software	£100	1	£100
Network Monitoring Software	£0 (open source)	-	£0

Subtotal Equipment Cost: £5,045

5.2 Installation and Cabling Costs

Assuming each employee has one wired connection, cabling is calculated as follows:

Item	Quantity	Unit Price	Total (£)
Ethernet Cables & Termination	100 runs	£100 per run	£10,000
AP and switch installation (labour)	3 floors	£300/floor	£900
Server rack setup	1 rack	£250	£250

Subtotal Installation Cost: £11,150

5.3 Ongoing Operational Costs (Annual)

Item	Cost (£/year)
Electricity & Cooling (Servers/APs)	£600
Maintenance/Support Staff (part-time)	£4,000
Domain & Email Hosting License (if hybrid)	£150
Backup Cloud Storage	£300

Annual Operational Subtotal: £5,050

5.4 Summary of Costs

Category	Total Cost (£)
Equipment	£5,045
Installation	£11,150
Initial Deployment	£16,195
Annual Operations	£5,050

6 Conclusion

This report presents a comprehensive and future-proof network proposal to assist TerraGaff Ltd. during its migration. After evaluating three network architectures, we have selected the Layer 3 routing model with centralized switching and VLAN partitioning as the most suitable one, mainly because of the improved security, scalability, and efficiency it offers. The design offers a separate subnet and VLAN on every other floor to maintain clean separation and uninterrupted data traffic. Managed switches, centralized DHCP, and VLAN-tagged wireless networks enable secure well-monitored access for both guests and personnel. Keeping essential services, such as email and the company website, on the premises brings about a couple of key benefits, namely, more authority over data and better business uptime. The structure of the building is established via a fast, wired backbone, which allows for future expansion as well. In summary, the proposed network not only fulfils the current demands adequately but is also ready for any future expansions.

Reference

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